0.1 Methodology: Step-by-Step Excel Implementation

The MP Materials (MP) Call option was valued in Excel using a One-Step Binomial Model calibrated on market and historical data.

Every step implementation can be checked in the provided file: `Report1_Quartucci_Davide.xlsx`

Step 1: Finding Data & Daily Returns Using Yahoo Finance, the Spot Price (S_0), the ATM Strike Price (K), and the Bid and Ask quotes of the selected 3-month and 6-month Call options were collected. In Excel, the market benchmark was computed as the Mid-Price:

$$P_{mid} = \frac{P_{bid} + P_{ask}}{2}$$

Step 2: Historical Data Extraction and Excel Integration To respect the *Matching Principle*, the volatility input was estimated on historical windows consistent with option maturity. Two datasets of daily closing prices (P_t) were downloaded from Yahoo Finance:

- A **3-month** historical sample.
- A **6-month** historical sample.

The CSV files were imported into Excel and stored in two separate worksheets, `3month` and `6month`, to keep the calculations distinct.

Step 3: Computation of Daily Returns In each Excel sheet, daily returns were computed from closing prices using:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t is the closing price on day t. In practice, a dedicated return column was created and filled down over the full sample.

Step 4: Estimation of Historical Volatility (σ_D and σ_Y) Historical volatility was estimated in Excel as the sample standard deviation of returns, using `=STDEV.S()` to obtain daily volatility (σ_D). This was then annualized as:

$$\sigma_Y = \sigma_D \cdot \sqrt{252}$$

The resulting values used in the pricing sheet were:

- **3-Month Sample:** $\sigma_D = 4,35\% \implies \sigma_Y = 69,07\%$
- **6-Month Sample:** $\sigma_D = 5,14\% \implies \sigma_Y = 81,63\%$

The longer lookback period captured higher historical variability, producing a larger volatility input for the 6-month model.

Step 5: Calibration of Binomial Price Multipliers (u and d) Using the CRR parametrization, the up and down multipliers were computed in Excel from annualized volatility and time to maturity (Δt):

$$u = e^{\sigma_Y \sqrt{\Delta t}}$$

$$d = e^{-\sigma_Y \sqrt{\Delta t}} = \frac{1}{u}$$

Excel formulas based on `EXP()` and `SQRT()` gave:

- **3-Month Horizon ($\sigma_Y = 69.07\%$):** $u = 1.4125$ and $d = 0.7080$
- **6-Month Horizon ($\sigma_Y = 81.63\%$):** $u = 1.7810$ and $d = 0.5615$

The higher 6-month volatility implies a wider range of possible future prices.

Step 6: Risk-Free Rate (SOFR) and Simple Compounding Factors The risk-free rate (r) was proxied by the 3-month and 6-month Term SOFR. In line with the project requirements, simple compounding was used instead of continuous compounding:

$$\text{Capitalisation Factor} = 1 + r\Delta t$$

$$\text{Discount Factor} = \frac{1}{1 + r\Delta t}$$

These expressions were implemented directly in Excel and linked to the pricing block.

Step 7: Risk-Neutral Probabilities and Theoretical Pricing (P_0) Having zero dividend yield, the risk-neutral probability was computed as:

$$q = \frac{(1 + r\Delta t) - d}{u - d}$$

The terminal Call payoffs were then evaluated in Excel with `MAX()`:

$$f_u = \max(S_0 u - K, 0)$$

$$f_d = \max(S_0 d - K, 0)$$

Finally, the theoretical price was obtained by discounting the expected payoff:

$$P_0 = \frac{1}{1 + r\Delta t}[q f_u + (1 - q) f_d]$$

The same Excel structure was replicated for both maturities, and the resulting P_0 values were compared with market Mid-Prices.

0.2 Visual Representation: The Binomial Tree Structure

Before examining the numerical results, it is instructive to visualize the geometry of the price process we have modeled. Figure 1 presents a stylized one-step binomial tree. The model assumes that from the current Spot Price (S_0), the asset price can only move to two possible states at maturity: an 'Up' state ($S_0 \cdot u$) and a 'Down' state ($S_0 \cdot d$).

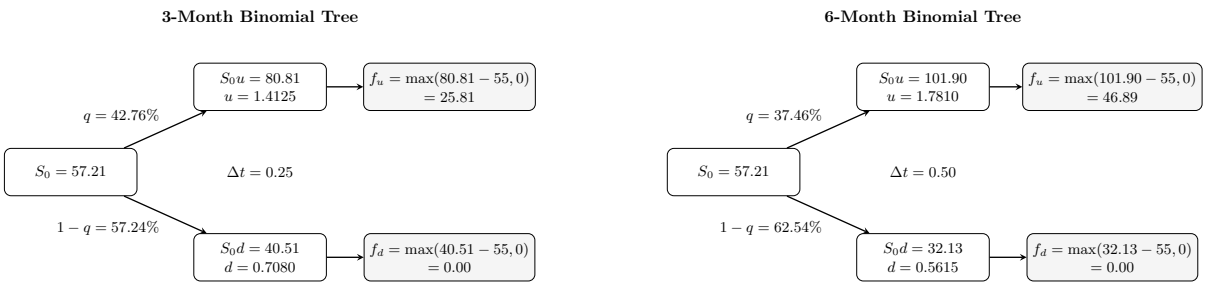

Figure 1: Personalized one-step binomial trees for the 3-month and 6-month MP Materials Call valuation in Excel.

0.3 Results Table & Critical Conclusions

The comparative analysis provides powerful insights into the mechanics of option pricing and the structural limitations of the One-Step Binomial Model when applied to high-volatility equities like MP Materials.

0.3.1 Confirmation of the Time Value Principle

The empirical results clearly demonstrate the fundamental principle of *Time Value*. By extending the option's maturity from 3 to 6 months, the theoretical premium (P_0) increases significantly from \$10.93 to \$17.25. A longer time horizon grants the underlying asset more time to experience favorable volatility, which is geometrically reflected in the massive increase of the Up-Node payoff (f_u), soaring from \$25.81 (3-month) to \$46.89 (6-month).

Parameter	3-Month Analysis	6-Month Analysis
Underlying & Market Data		
Spot Price (S_0)	\$57.21	\$57.21
Strike Price (K)	\$55.00	\$55.00
Market Bid / Ask	\$9.45 / \$9.60	\$12.25 / \$13.25
Market Mid-Price	**\$9.53**	**\$12.75**
Model Inputs		
Annualized Volatility (σ_Y)	69.07%	81.63%
Risk-Free Rate (SOFR)	3.68%	3.66%
Capitalisation Factor	1.0092	1.0183
Discount Factor	0.9909	0.9820
Binomial Dynamics		
Up Factor (u) / Down Factor (d)	1.4125 / 0.7080	1.7810 / 0.5615
Risk-Neutral Prob. (q / $1-q$)	42.76% / 57.24%	37.46% / 62.54%
Payoff Up (f_u) / Payoff Down (f_d)	\$25.81 / \$0.00	\$46.89 / \$0.00
Valuation & Spread		
Theoretical Price (P_0)	**\$10.93**	**\$17.25**
Pricing Gap (Model Spread \$)	+\$1.41	+\$4.50
Pricing Gap (Model Spread %)	**+14.79%**	**+35.30%**

Table 1: Consolidated Output: One-Step Binomial Model vs. Market Pricing.

0.3.2 Divergence: Historical vs. Implied Volatility

The most critical finding of this report is the systematic overvaluation of the options by the theoretical model, characterized by a positive pricing gap of +14.79% for the 3-month contract and a substantial +35.30% for the 6-month contract.

This discrepancy highlights the fundamental difference between backward-looking and forward-looking risk metrics. The model was calibrated using **Historical Volatility** (σ_Y of 69.07% and 81.63%), capturing a period of extreme market turbulence for MP Materials (evident from its wide 52-week range). However, the real market Mid-Prices (\$9.53 and \$12.75) are driven by **Implied Volatility**. The lower market premiums indicate that institutional investors and market makers expect the future price action of MP Materials to stabilize, thereby pricing in a lower volatility regime than what was observed in the past.

0.3.3 Structural Limitations of the One-Step Framework

Finally, the widening of the error margin in the 6-month analysis (+35.30%) exposes the inherent limitations of a single-period binomial tree. Oversimplifying 6 months of continuous trading into a single discrete jump creates extreme, unrealistic terminal states. Specifically, the model assumes the stock will either surge by 78.1% ($u = 1.7810$) or crash by 43.8% ($d = 0.5615$) in exactly one step. In reality, option pricing over longer horizons requires multi-step lattices (e.g., 50 or 100 steps) or continuous-time models (Black-Scholes) to properly smooth the distribution of terminal prices and align theoretical values closer to market realities.